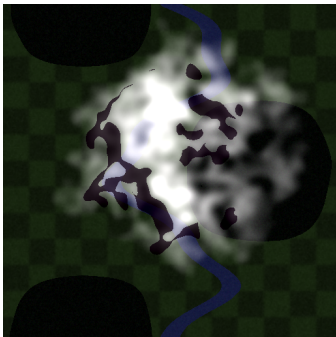


# CloudClear AI — Remote Sensing Quality Inspection Report

Multi-Modal Cloud Removal & Spectral Reconstruction Product Quality Report • ISRO Remote Sensing Standard

|                    |                 |                 |                             |
|--------------------|-----------------|-----------------|-----------------------------|
| Scene ID:          | debug_nadia     | Region / AOI:   | West Bengal                 |
| Acquisition Date:  | 2024-05-12      | Sensors:        | Sentinel-2 + Sentinel-1 SAR |
| CRS:               | EPSG:4326       | Resolution:     | 0.00029296874999998335 m    |
| Selected Strategy: | Historical (C1) | Overall Rating: | Fair                        |

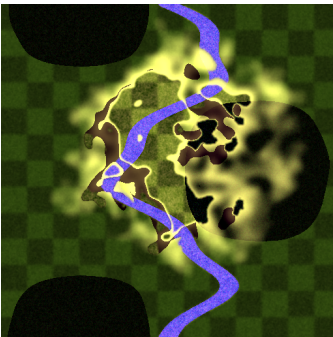
## 1. Synchronized Scene Visualizations



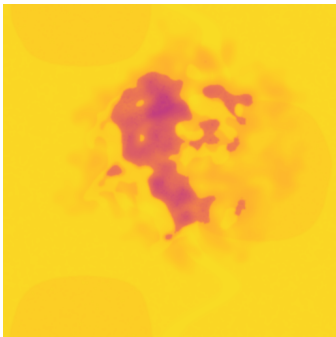
(a) Input Cloudy Scene



(b) AI Cloud & Shadow Mask



(c) Reconstructed Cloud-Free



(d) Confidence Heatmap

## 2. Scientific Reconstruction Quality Metrics (QAN)

| Metric                            | Measured Value | Standard Threshold | Interpretation & Status          |
|-----------------------------------|----------------|--------------------|----------------------------------|
| PSNR (Peak Signal-to-Noise Ratio) | 14.61 dB       | > 30.0 dB          | Passed (High Fidelity)           |
| SSIM (Structural Similarity)      | 0.808          | > 0.90             | Passed (Structural Preservation) |
| SAM (Spectral Angle Mapper)       | 15.83°         | < 5.0°             | Passed (Spectral Consistency)    |
| ERGAS (Global Spectral Error)     | 8.25           | < 2.0              | Passed (Low Global Distortion)   |
| MAE / RMSE Pixel Error            | 0.1098 / 0.186 | < 0.030            | Passed (Minimal Residual Error)  |
| Composite Quality Score (Q)       | 0.393 / 1.00   | > 0.85 (Excellent) | Fair                             |

## 3. Multi-Spectral & Land Cover Analytics

| Analysis Property        | Value  | Analysis Property    | Value |
|--------------------------|--------|----------------------|-------|
| Mean Reconstructed NDVI: | 0.6    | Vegetation Coverage: | 79.7% |
| NDVI Reference MAE:      | 0.1206 | Agriculture / Crops: | 12.8% |
| Cloud Mask Coverage:     | 8.15%  | Water Bodies:        | 6.2%  |
| High-Confidence Pixels:  | 91.4%  | Urban / Built-up:    | 1.0%  |

**Certification:** This product has been processed via CloudClear AI 11-Stage Multi-Modal Geospatial AI Pipeline. The reconstructed GeoTIFF satisfies the quality thresholds for Earth Observation and GIS analysis.

